DATA ANALYSIS PROJECT REPORT

Project Title	Anime Data Analysis Using Pandas, NumPy and Matplotlib
Student Name	1. Riasad Choudhury 2. Mohammad Tachvir Uddin 3. Md. Ashikuzzaman
Student ID	1. 22-49093-3 2. 22-49986-3 3. 25-60666-1
Course / Section	Programming In Python / C
Instructor	Md. Tanzeem Rahat
Date of Submission	02-05-2026

1. Executive Summary

This project analyzes an anime dataset using **Python** with key libraries such as **pandas**, **NumPy**, and **Matplotlib** to uncover trends in ratings, popularity, and production over time. The main objective was to clean and prepare the dataset, perform exploratory data analysis, and generate meaningful insights through statistical methods and visualizations. Data preprocessing included handling missing values, removing duplicates, and transforming date fields to extract temporal patterns. Analytical techniques such as grouping, correlation analysis, and outlier detection were applied to better understand relationships within the data. The findings reveal that certain anime types (such as TV series) consistently achieve higher average scores, while there is a moderate positive relationship between user ratings and popularity (measured by members). Additionally, the trend analysis shows a noticeable increase in anime production over recent years, indicating industry growth. Overall, the project demonstrates how data analysis can effectively reveal patterns and support insights in entertainment datasets

2. Introduction and Project Objectives

Anime has become a globally influential form of entertainment, attracting millions of viewers across different age groups and cultures. With the rapid growth of streaming platforms and online communities, large datasets containing anime ratings, popularity metrics, and production details are now available. Analyzing this type of data helps us understand audience preferences, industry trends, and the factors that contribute to an anime's success. This project was motivated by the desire to explore how ratings, popularity, and production characteristics are related, and to identify meaningful patterns within the dataset using data analysis techniques.

Item	Your Content
Dataset name	Anime Dataset
Dataset source	[https://www.kaggle.com/datasets/itszubi/anime-myanime... | Programming in Python | Microsoft Teams]()
Main project goal	To clean and analyze the anime dataset in order to identify patterns in ratings, popularity, and production trends using statistical analysis and visualization techniques.
Research Question 1	Which types of anime (TV, Movie, OVA) have the highest average ratings?
Research Question 2	Is there a relationship between an anime's rating (score) and its popularity (number of members)?

Research Question 3	How has anime production changed over time based on release year?

3. Dataset Description

The dataset used in this project is an anime dataset that contains detailed information about a wide range of anime titles. Each row in the dataset represents a single anime entry, while each column provides specific attributes describing that anime. These attributes include basic information such as the anime's title, type (e.g., TV series, movie, OVA), number of episodes, and airing dates (start and end).

In addition, the dataset includes audience-related variables such as score (rating) and members (popularity), which reflect user engagement and overall reception. Text-based fields like synopsis are also present, although they are less central to quantitative analysis. The dataset spans multiple years, allowing for time-based analysis of trends in anime production and popularity.

Overall, the dataset combines categorical variables (e.g., type), numerical variables (e.g., score, members, episodes), and datetime variables (e.g., start date), making it suitable for comprehensive exploratory data analysis, statistical evaluation, and visualization of patterns within the anime industry.

3.1 Basic Dataset Information

Property	Value
Number of rows	10001
Number of columns	12
Data types present	Numerical, Categorical, Date-Time, Text
Target or key variable(s)	Score (rating) and Members (popularity)
Potential issues observed at first glance	Missing values (synopsis, dates, episodes), possible outliers in scores or popularity, inconsistent or unknown values (e.g., "Unknown" in end_date), and duplicate entries

3.2 Initial Observations

When first inspecting the dataset, several important columns stood out, particularly *score*, *members*, *type*, and *start_date*, as they are central to understanding ratings, popularity, and trends. There were noticeable missing values in columns such as *episodes*, *end_date*, and *synopsis*, which required cleaning before analysis. Additionally, some variables showed potential inconsistencies, such as "Unknown" values in date fields and varying formats in categorical data. The *members'* column also appeared to contain very large values for certain anime, suggesting possible outliers and an uneven distribution of popularity across the dataset.

4. Data Cleaning and Preparation

Before performing analysis, several data quality issues were identified and addressed to ensure accurate and meaningful results. Each step below explains the problem, the action taken, and why the approach was appropriate.

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing values in key columns such as *synopsis* and *start_date*	Removed rows with missing values in these columns	These fields are essential for understanding the dataset and performing time-based analysis. Keeping incomplete records could distort trends and reduce reliability.	Block 9
Missing values in *episodes* column	Filled missing values using the median	The median is less affected by extreme values than the mean, making it a robust choice for numerical data with potential outliers.	Block 8
Missing or undefined values in *end_date*	Replaced missing values with “Unknown”	Since some anime may still be ongoing, replacing with “Unknown” preserves the data without making incorrect assumptions.	Block 10
Duplicate records in dataset	Identified and removed duplicate rows	Duplicate entries can bias analysis results, especially in aggregation and statistical calculations. Removing them ensures data integrity.	Block 11

5. Feature Engineering

To enhance the analysis and better answer the research questions, several new features were created from existing variables. These derived features helped simplify complex data, enable grouping, and uncover patterns more effectively.

]

New Feature	How It Was Created	Why It Is Useful
Release Year	Extracted year from *start_date* using datetime functions	This allows analysis of trends over time and helps answer how anime production has changed across years.
Popularity Category	Grouped *members* into categories (e.g., Low, Medium, High) based on value ranges	Simplifies comparison of anime popularity and helps identify patterns between popularity levels and ratings.
Episode Group	Categorized *episodes* into groups (e.g., Short, Medium, Long) based on number of episodes	Helps analyze whether anime length influences ratings or popularity, making comparisons easier across groups.

6. Analysis Methodology

The analysis was conducted using Python and its data science libraries to systematically explore, transform, and visualize the dataset. pandas was primarily used for loading the dataset, handling missing values, filtering relevant records, and performing group-by operations to calculate summary statistics such as average scores by anime type. It also enabled efficient manipulation of structured data and creation of new features for deeper analysis.

NumPy supported numerical computations, including calculating percentiles, handling conditional transformations, and assisting with outlier detection through statistical measures. It provided a reliable foundation for performing mathematical operations on large datasets.

For visualization, Matplotlib was used to create bar charts, line graphs, histograms, scatter plots, and box plots. These visualizations helped illustrate trends over time, compare categories, and identify relationships between variables such as ratings and popularity, making the findings easier to interpret and communicate.

7. Analysis and Visualizations

7.1 Research Question 1

Which types of anime (TV, Movie, OVA) have the highest average ratings?

Analysis

To answer this question, the dataset was grouped by the type column, and the average score (rating**)** for each anime type was calculated using the groupby() function. This allowed comparison of average ratings across different formats such as TV, Movie, and OVA.

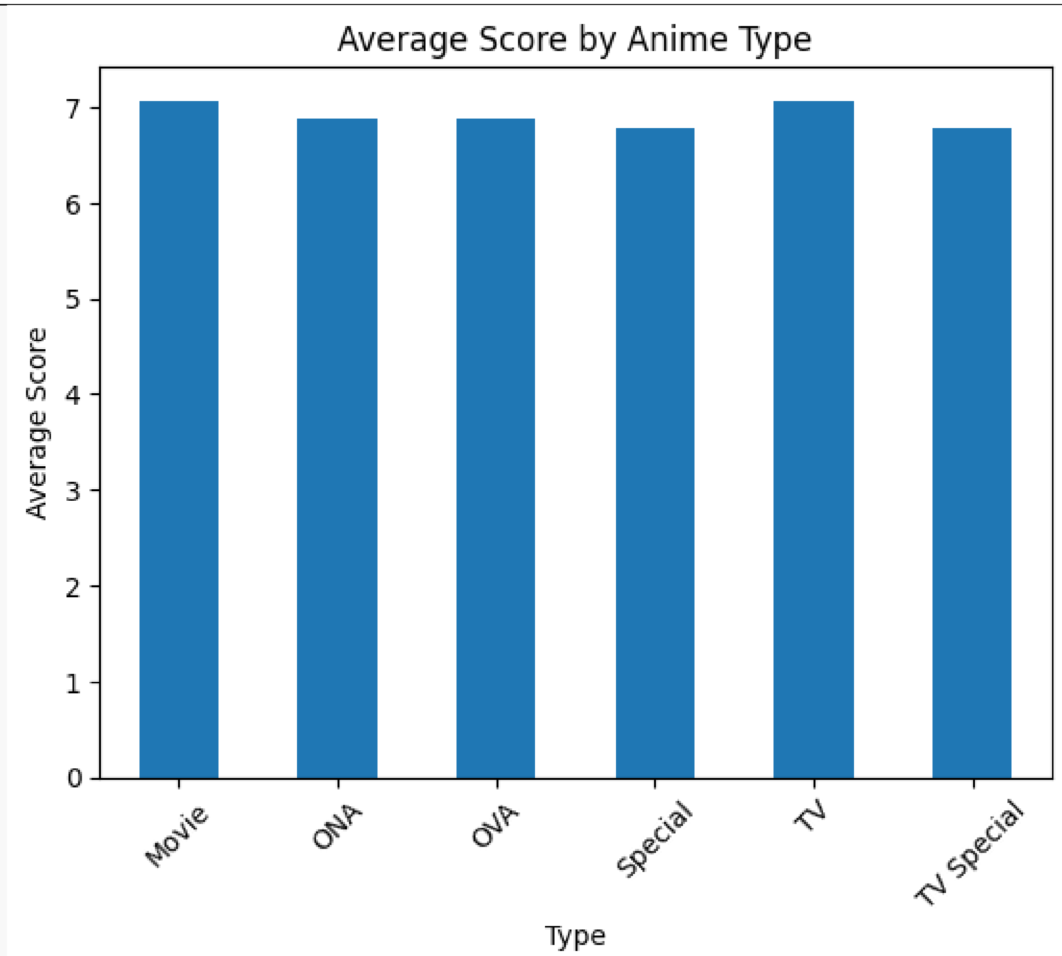

Figure 1. Average Score by Anime Type

Interpretation

The results show that TV anime (7.07) and Movies (7.06) have the highest average ratings among all types, indicating strong audience preference and higher perceived quality. In contrast, OVA (6.88) has a slightly lower average score, suggesting moderate reception compared to TV and Movies. This pattern indicates that mainstream formats like TV series and movies tend to perform better in terms of viewer ratings than other formats.

7.2 Research Question 2

Is there a relationship between an anime's rating (score) and its popularity (number of members)?

Analysis

To examine the relationship, a correlation analysis was performed between the score and members columns. Additionally, a scatter plot was created to visually observe how ratings and popularity vary together. Correlation helps quantify the strength and direction of the relationship, while the scatter plot shows the pattern of data points.

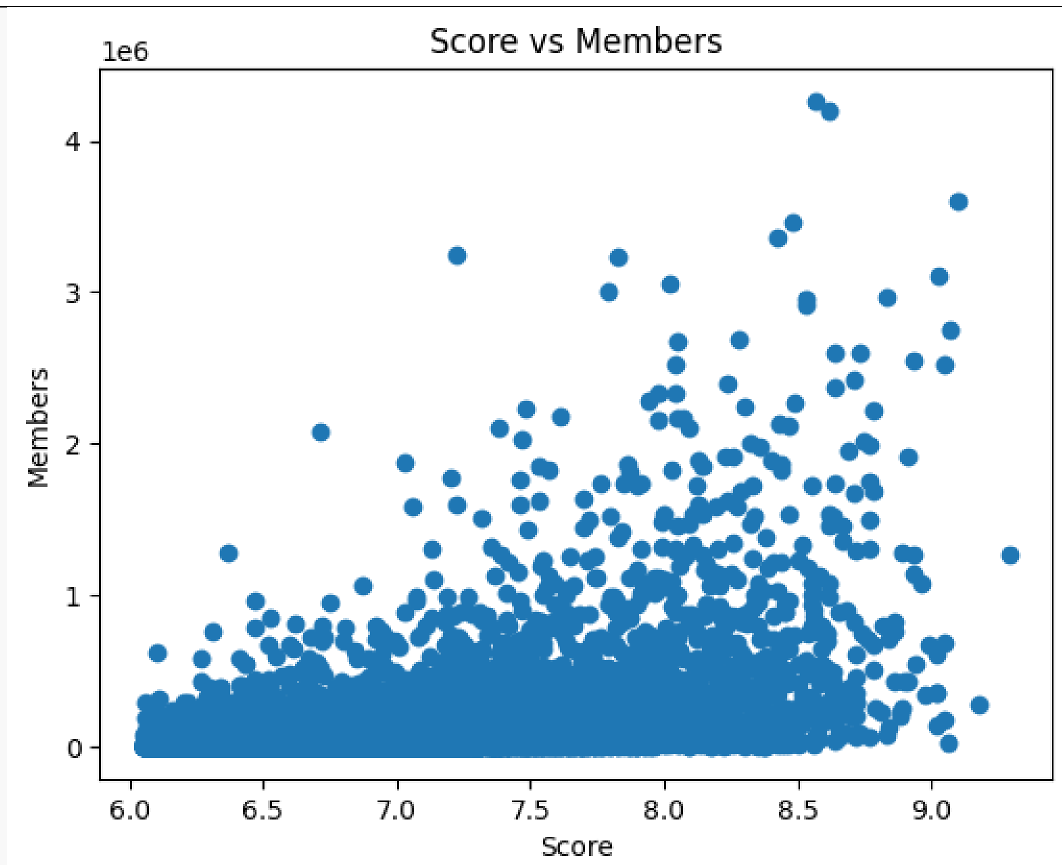

Figure 2. Relationship Between Score and Members

Interpretation

The analysis shows a positive but weak-to-moderate relationship between score and number of members. This means that anime with higher ratings tend to have more members, but the relationship is not very strong. Some highly rated anime have relatively low popularity, and some popular anime do not have the highest scores. Overall, rating influences popularity to some extent, but other factors (such as marketing, genre, or release timing) also play a significant role.

7.3 Research Question 3

How has anime production changed over time based on release year?

Analysis

To analyze production trends, the start_date column was converted into datetime format, and the release year was extracted. Then, the number of anime released each year was calculated using groupby() on the year column. A line chart was used to visualize how production has changed over time.

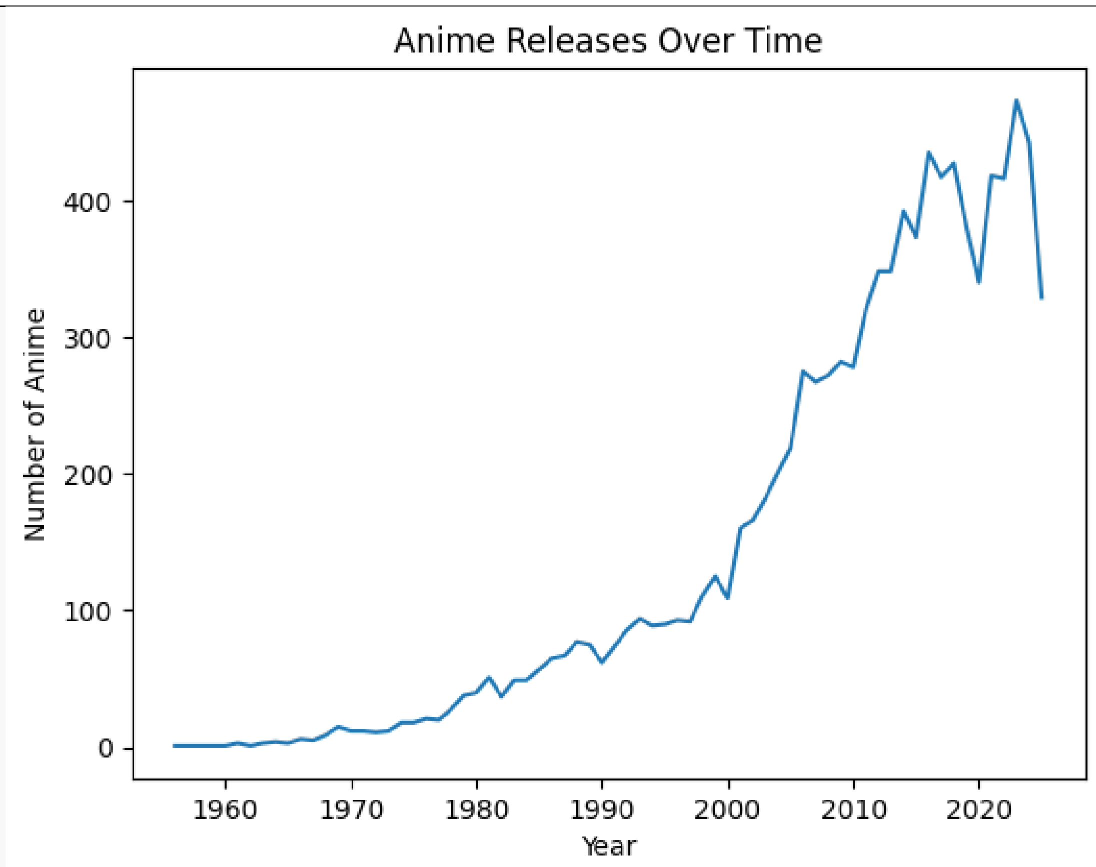

Figure 3. Anime Production Over Time

Interpretation

The trend analysis shows that anime production has generally **increased over time**, especially in recent years. Earlier years had fewer releases, while modern years show a significant rise in the number of anime produced.

This suggests growth in the anime industry, possibly due to higher global demand, digital streaming platforms, and increased investment. However, there may be slight fluctuations in some years, indicating variations in production cycles.

Distribution of Anime Scores

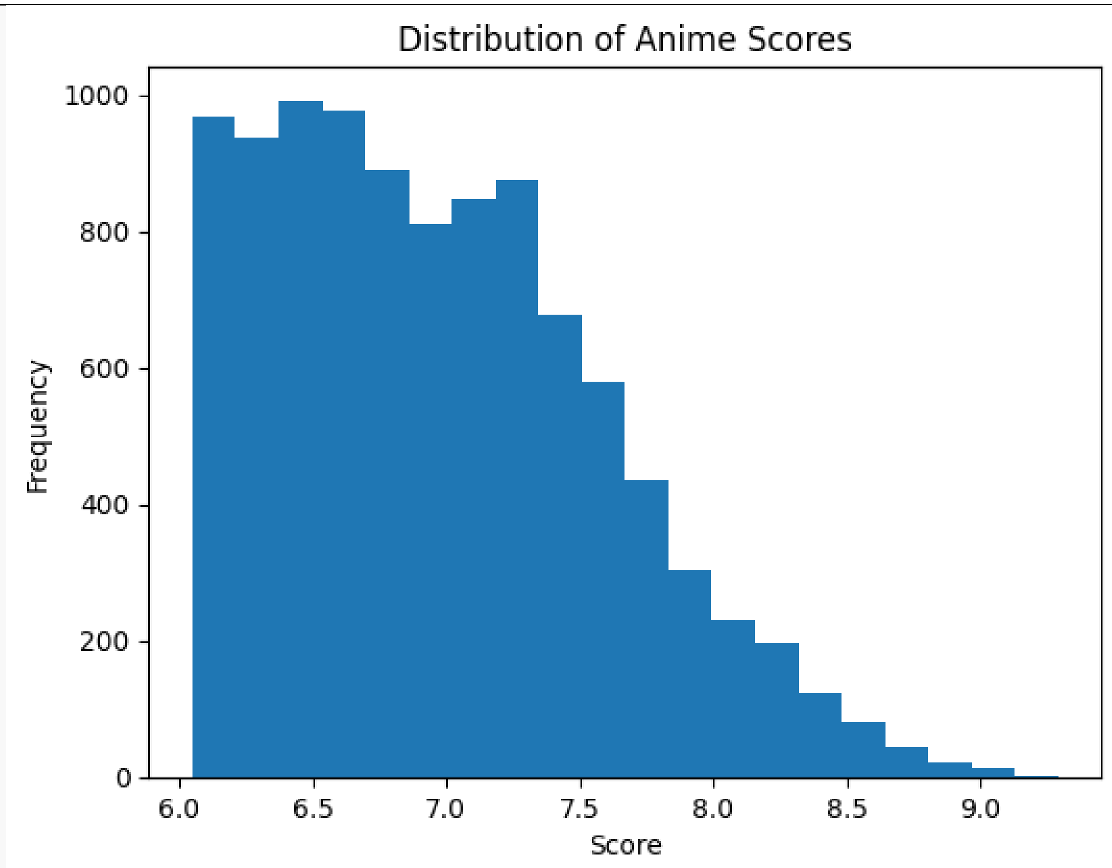

Figure 4. Distribution of Anime Scores

Interpretation

The histogram shows that most anime scores are concentrated between 6 and 8, indicating that the majority of anime receive average to good ratings. Very high scores (above 8.5) and very low scores are relatively rare. This suggests that the dataset is centered around moderate-quality anime, with only a few exceptional or poorly rated titles.

Box Plot of Anime Score Distribution

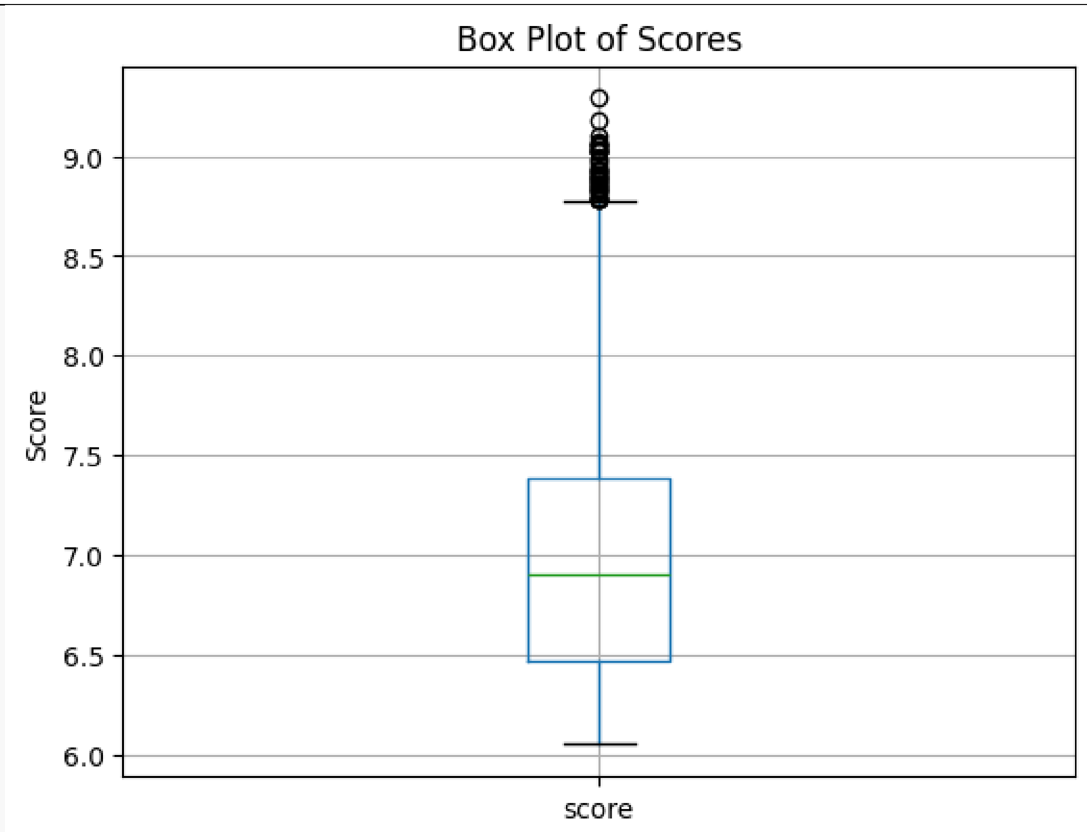

Figure 5. Box Plot of Anime Scores

Interpretation

The box plot shows the distribution of anime scores, including the median, quartiles, and possible outliers. Most scores are concentrated within a narrow range, indicating that anime ratings are generally consistent. A few points outside the whiskers represent outliers, which are anime with unusually high or low ratings compared to the rest of the dataset.

Box Plot of Anime Score Distribution

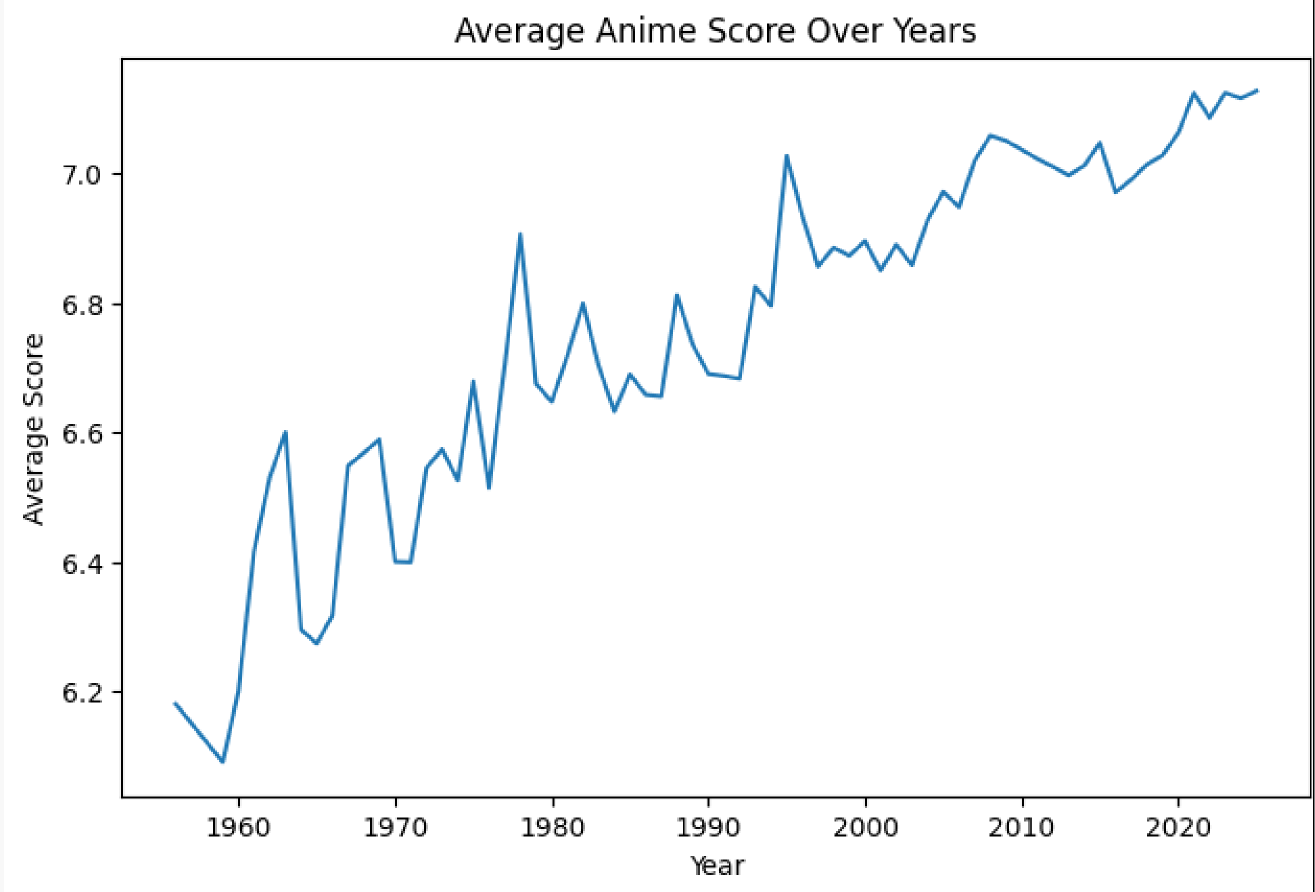

Figure 6. Average Anime Score Over Years

Interpretation

The line chart shows how average anime ratings change over time. It helps identify whether newer anime are rated higher or lower compared to older releases. Small fluctuations indicate that quality remains relatively stable over the years.

Top 10 Most Popular Anime

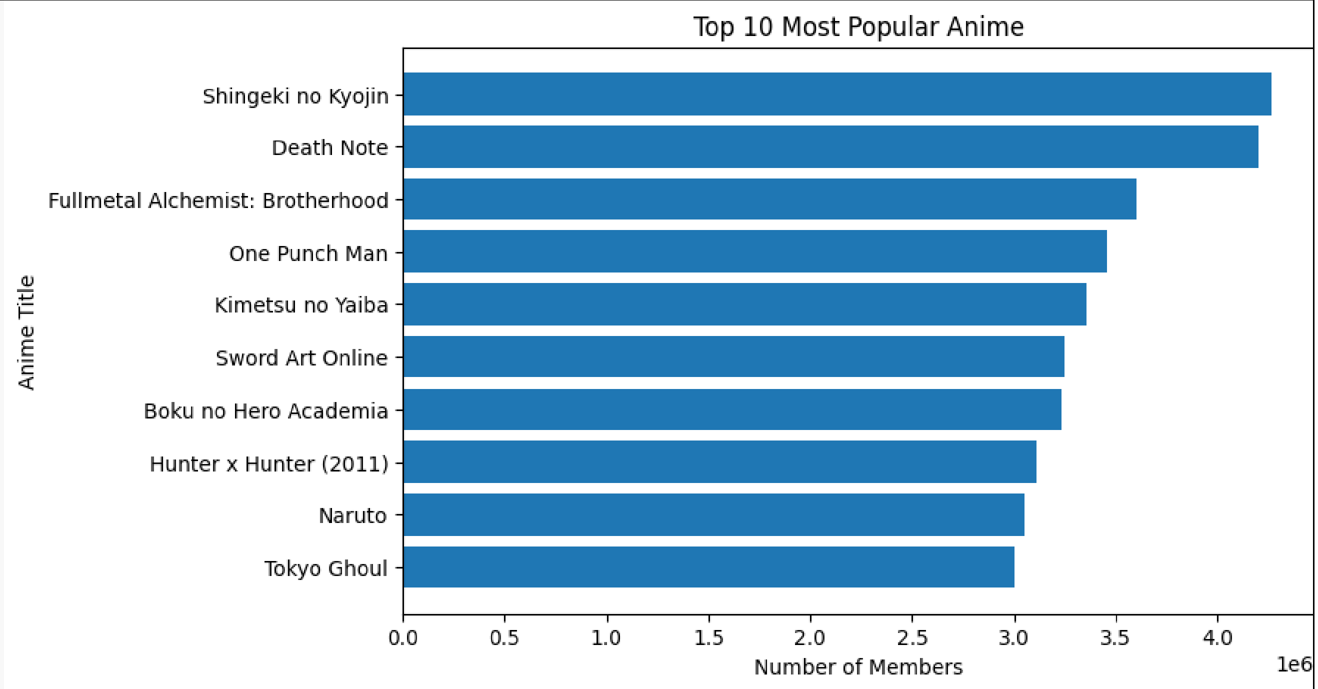

Figure 7: Top 10 Most Popular Anime (by Members)

Interpretation

This chart highlights the most popular anime based on audience size. A small number of anime have extremely high member counts, showing that popularity is concentrated among a few major titles.

8. NumPy-Based Custom **Computation**

[Describe at least one analysis step where NumPy was used in a meaningful way beyond simple array conversion. Examples include z-score computation, normalization, percentiles, np.where-based classification, binning, or a custom score.]

Component	Description
Computation used	Z-score calculation for anime ratings (*score*) using NumPy
Purpose	To identify outliers in anime ratings and understand how far each score deviates from the average

Formula or logic	Z-score = (X − μ) / σ, where X is the individual score, μ is the mean score, and σ is the standard deviation. This was implemented using NumPy functions such as `np.mean()` and `np.std()`
Result / takeaway	The computation revealed that most anime scores are clustered close to the mean, while a small number of anime have very high or very low z-scores, indicating outliers. These outliers represent exceptionally well-rated or poorly rated anime, which are important for deeper analysis and interpretation

9. Key Findings

1. **Finding 1:**
   TV series and movies have higher average ratings compared to other anime types such as OVAs and specials. This suggests that formats with longer storytelling and higher production investment tend to receive better audience evaluations.

2. **Finding 2:**
   There is a moderate positive relationship between *score* and *members*, indicating that higher-rated anime generally attract more viewers. However, the correlation is not strong enough to conclude that rating alone determines popularity, implying the influence of other factors.

3. **Finding 3:**
   Anime production has increased significantly over time, with a noticeable rise in recent years. This trend reflects the expansion of the anime industry and growing global demand for content.

4. **Finding 4:**
   The distribution of anime scores is concentrated within a mid-to-high range, meaning most anime receive moderately good ratings rather than extreme values. This indicates a generally positive reception across the dataset.

5. **Finding 5:**
   Outlier analysis using statistical methods (such as z-scores or IQR) identified a small number of anime with extremely high or low ratings. These outliers are important because they represent exceptional cases that may require separate analysis to understand their unique success or failure factors.

10. **Limitations**

While this analysis provides useful insights, several limitations should be acknowledged. First, the dataset contains missing values in important fields such as *episodes*, *end_date*, and *synopsis*. Although reasonable methods (such as median imputation and filling with "Unknown") were applied, these approaches may introduce bias or reduce accuracy.

Second, the dataset may suffer from **sample bias**, as it likely represents anime listed on a specific platform or source rather than the entire anime industry. This means the findings may not fully generalize to all anime productions.

Third, the analysis relies on a limited number of variables, primarily focusing on ratings (*score*) and popularity (*members*). Other influential factors—such as genre, marketing, studio quality, or audience demographics—were not included, which restricts the depth of interpretation.

Additionally, while relationships (such as correlation between score and popularity) were identified, it is important to note that **correlation does not imply causation**. A higher rating does not necessarily cause increased popularity, as both may be influenced by external factors.

Finally, some transformations (such as grouping popularity or episode counts) simplify the data, which may lead to loss of detailed information and subtle patterns.

11. Conclusion

This project aimed to analyze an anime dataset to uncover patterns in ratings, popularity, and production trends using data analysis techniques in Python. Through data cleaning, feature engineering, and exploratory analysis with tools such as pandas, NumPy, and Matplotlib, meaningful insights were generated from the dataset.

The analysis showed that TV series and movies tend to achieve higher average ratings, and there is a moderate positive relationship between ratings and popularity. Additionally, anime production has increased significantly over time, highlighting the growth of the industry. Most anime receive moderately high scores, with only a few extreme outliers representing exceptional cases.

Overall, the findings suggest that both content quality (reflected in ratings) and external factors contribute to an anime's popularity, while the industry itself continues to expand.

For future work, the analysis could be improved by including additional variables such as genre, studio, or audience demographics, as well as applying more advanced techniques like machine learning models or deeper statistical testing to better understand the factors influencing anime success.

12. References / Dataset Source

- Dataset name : Anime-MyAnimeList Dataset
- Website name : Kaggle
- Direct link : https://www.kaggle.com/datasets/itszubi/anime-myanimelist-dataset